

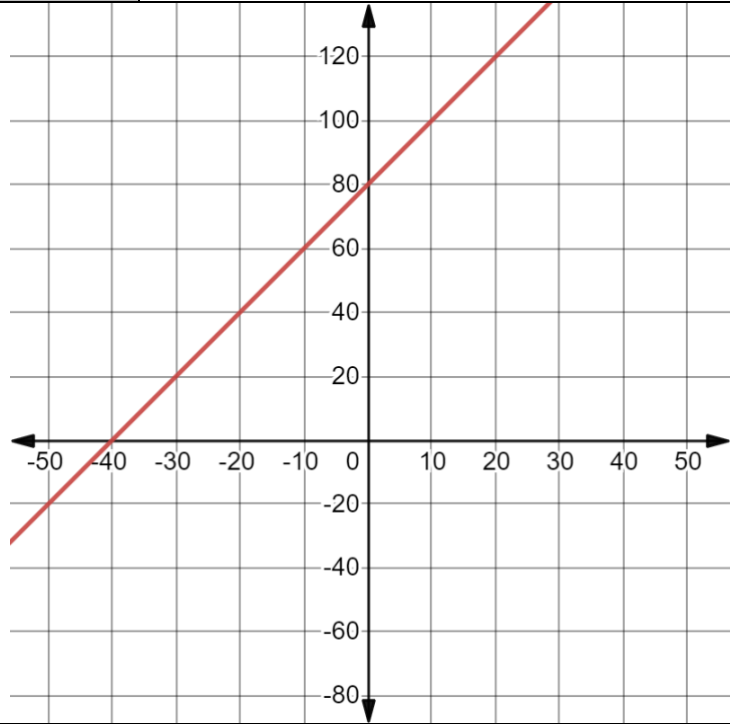
Algebra 1 Unit 4 Assessment Guide

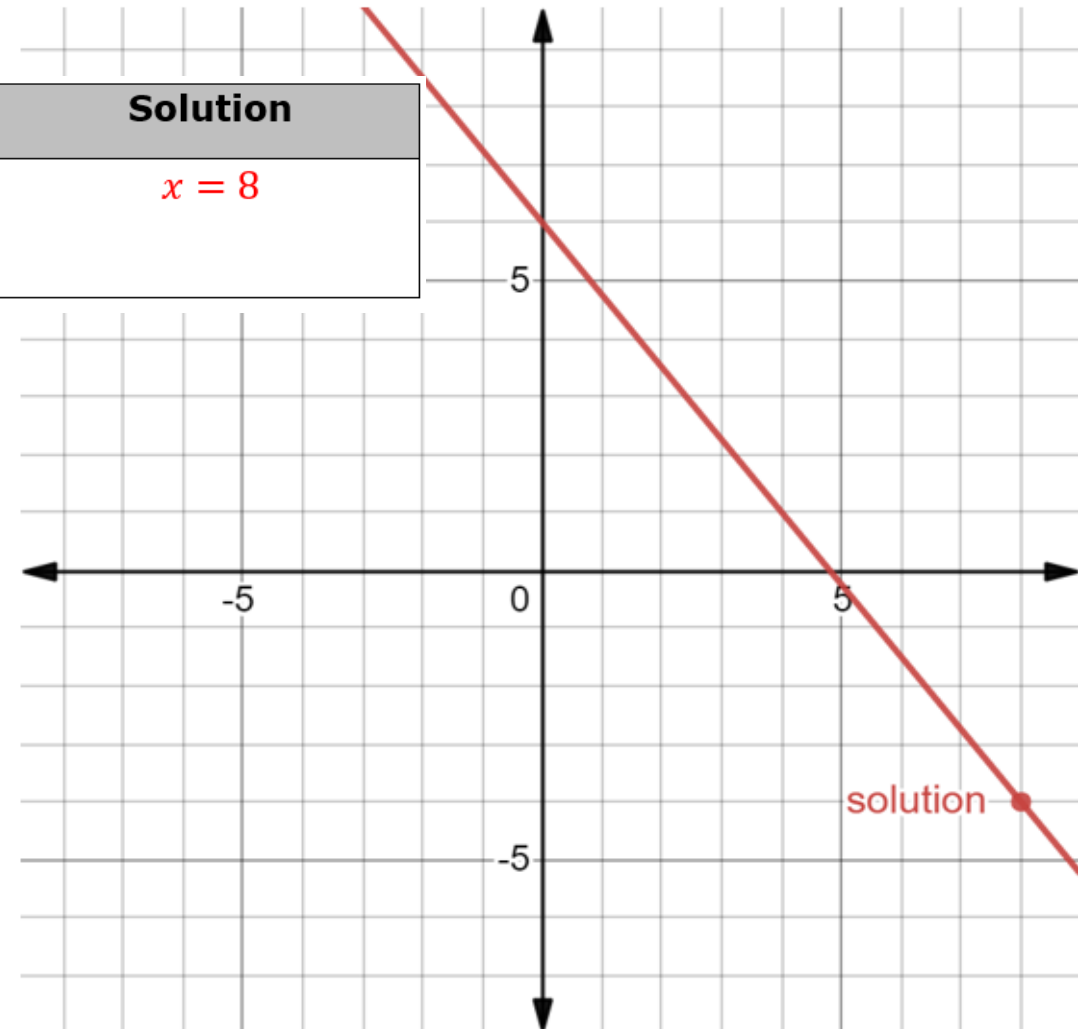
General Information	The questions on the unit assessment are calculator neutral, which means that a calculator does not give a student an advantage or disadvantage when showing their knowledge of the intended benchmark. It is suggested that a student is provided a scientific calculator for this assessment.
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Question	1	Benchmark(s)	MA.912.F.1.5						
<table><tr><td>Inequality</td><td>Set-Builder Notation</td></tr><tr><td>$x \geq -8$</td><td>$\{x: x \geq -8\}$</td></tr></table>		Inequality	Set-Builder Notation	$x \geq -8$	$\{x: x \geq -8\}$	Recommended Scoring	Each answer should be considered as separate responses. Suggested points for each is 3 points.		
Inequality	Set-Builder Notation								
$x \geq -8$	$\{x: x \geq -8\}$								
Question	2	Benchmark(s)	MA.912.AR.2.4						
<table><tr><td>Key Feature</td><td>End Behavior</td><td>y-intercept</td></tr><tr><td>Description</td><td>As x-values increase, y-values <i>decrease</i>.</td><td>The y-intercept is <i>(0,5)</i>.</td></tr></table>		Key Feature	End Behavior	y-intercept	Description	As x-values increase, y-values <i>decrease</i> .	The y-intercept is <i>(0,5)</i> .	Recommended Scoring	Each answer should be considered as separate responses. Suggested points for each is 5 points. Accept 5 for the y-intercept.
Key Feature	End Behavior	y-intercept							
Description	As x-values increase, y-values <i>decrease</i> .	The y-intercept is <i>(0,5)</i> .							

Question	3	Benchmark(s)	MA.912.AR.2.3
Point-Slope Form $y + 7 = \frac{5}{2}(x - 4)$ Slope-Intercept Form $y = \frac{5}{2}x - 17$	Recommended Scoring	Each answer should be considered as separate responses. Suggested points for each is 5 points.	
	Additional Notes	Consider student's work for partial credit. Consider giving students who have the correct slope for each form at total of 4 points.	
Question	4	Benchmark(s)	MA.912.AR.2.3
$x = -3$	Recommended Scoring	Suggested points is 3 points.	
	Additional Notes	This is commonly missed by students.	

Question	5	Benchmark(s)	MA.912.AR.2.3												
Standard form of parallel line <u>$-3x + y = -10$</u>	Recommended Scoring	Suggested points is 5 points. Accept the equivalent form $3x - y = 10$ as correct.													
	Additional Notes	Consider student work for partial credit. If students show a correct form at some point in their work and an incorrect standard form, then consider subtracting 2 points. If it appears that students incorrectly determined slope from the graph but all other work is correct, consider taking off 1 point.													
Question	6a	Benchmark(s)	MA.912.AR.2.5												
<table><tr><th>Domain</th><th>Range</th><th>As x-values increase, y-values ...</th></tr><tr><td>$0 \leq x \leq 4$</td><td>$80 \leq x \leq 88$</td><td><i>increase</i></td></tr><tr><th>Slope</th><th>x-intercept</th><th>y-intercept</th></tr><tr><td>2</td><td>$(-40,0)$</td><td>$(0,80)$</td></tr></table>	Domain	Range	As x -values increase, y -values ...	$0 \leq x \leq 4$	$80 \leq x \leq 88$	<i>increase</i>	Slope	x -intercept	y -intercept	2	$(-40,0)$	$(0,80)$	Recommended Scoring	Suggested points is 4 points for each key feature.	
	Domain	Range	As x -values increase, y -values ...												
	$0 \leq x \leq 4$	$80 \leq x \leq 88$	<i>increase</i>												
	Slope	x -intercept	y -intercept												
	2	$(-40,0)$	$(0,80)$												
Additional Notes	The domain and range will be the most commonly missed answers. Consider partial credit for a student who determines the domain and range using 2013 and 2017. For slope accept equivalent forms. For x -intercept accept -40 . For y -intercept accept 80.														

Question	6b	Benchmark(s)	MA.912.AR.2.4
	Recommended Scoring	Suggested points is 5 points.	
	Additional Notes	Look for students to use the x - and y -intercepts. Students who attempt to use slope will have an incorrect line unless they considered the values. Suggest giving 2 points to students who have the correct y -intercept but who did not correctly use the scale for the slope.	

Question	7	Benchmark(s)	MA.912.AR.2.4
Solution $x = 8$ 	Recommended Scoring	Suggested points is 3 points for the correct graph, 2 points for the correct solution labeled on the graph and 3 points for the solution written as $x = 8$ or simply 8.	
	Additional Notes	Students should not give the ordered pair as the solution.	

Question	8e	Benchmark(s)	MA.912.AR.2.5
Reasonable? ☐ No ☐ Yes Explanation <i>Answers will vary: The x-intercept is approximately $(-16, 0)$, which means the year would be 1983 which is not within the given time period.</i>		Recommended Scoring	Suggested points is 2 points for no and 2 points for correct explanation.
		Additional Notes	Students may say it is not reasonable because time can't be negative. In this case, a negative represents 16 years before the time period so time can be negative in this case.
Question	9	Benchmark(s)	MA.912.F.1.5
Select the phrases that complete the comparison of the key features. The x-intercept for ☐ line A ☒ line B is greater than the x-intercept for ☒ line A. ☐ line B. The range for line A is ☒ the same as ☐ different than the range for line B. The slope for ☐ line A ☒ line B is smaller than the slope for ☒ line A. ☐ line B.		Recommended Scoring	Suggested points is 3 points for each statement.